SOUTH LAKE HOME EDUCATORS



Lesson 14

Raising the Stakes

Semester 1, Week 14 - Difficulty Scaling & Stretch Features

At a Glance

Lesson 14 objectives & materials

By the end of today, students will:

- Replace hardcoded numbers with wave-dependent formulas
- Keep every referenced value in agreement, everywhere
- Scale both enemy count AND enemy speed by wave
- Choose and attempt one capstone-level stretch feature

What We're Using Today

- Demo laptop with all 8 engines verified
- GameMaker, for the live lecture demo
- No new assets - reuses existing enemies/spawn logic
- 001GC: 2 new pre-placed enemy copies (design time)
- Your own [YourName]Game project from Lesson 13

IGDA framework tie-in: Game Design - balance, pacing, difficulty curves (intermediate); Programming - parameterization (intermediate).

Today's Agenda



0:00 - 0:05	Welcome back, Lesson 13 recap
0:05 - 0:15	The Foreshadowed Feature: One Formula, Two Places
0:15 - 0:35	Adding Speed to the Mix
0:35 - 0:45	This Week's Twist: Student's Choice
0:45 - 0:55	Introduce lab exercise, hand out engine-specific handouts
LAB TIME	***— Raising the Stakes —***
0:55 - 1:50	Scale count and speed, then pick one stretch feature
1:50 - 2:00	Exit check: toughest wave 3 wins bragging rights



Welcome Back

Today, your game starts fighting back.

5 min

Welcome Back

5 min

Quick recap of Lesson 13:

- Getting hit finally looks like getting hit - flash, then explosion
- Lesson 11's Key deliberately called 5-per-wave a "simplification"
- Today's the payoff: count AND speed both scale with wave

Open your own project from last week today, not a new one. No new assets needed - 001 Game Creator students add two more pre-placed enemy copies first (see your Handout).



One Formula, Two Places

The payoff for a simplification flagged weeks ago.

10 min

One Formula, Two Places

10 min

Instead of always spawning 5 enemies, spawn (4 + wave). Wave 1: still 5. Wave 2: 6. Wave 3: 7. One small formula, and the game already escalates.

The catch: that formula has to appear in TWO places - wherever enemies spawn, AND wherever enemies_remaining resets for a new wave. If they disagree, the win-check breaks.

One source of truth, referenced everywhere - the same discipline behind Lesson 10's reusable reset function.



Adding Speed to the Mix

Count is a start. Speed makes it dangerous.

20 min

Adding Speed to the Mix

20 min

Count alone is a good start, but speed makes it feel dangerous.

GameMaker demo: in spawn_enemies(), right after creating each enemy instance, set inst.enemy_speed = 1 + (wave - 1) * 0.5. Wave 1 stays the same; wave 3 is noticeably faster.

That new enemy_speed variable has to actually get used: obj_enemy's Step event currently just says y += 1, so change that one line to y += enemy_speed. Notice what did NOT change: the rest of the Step event, from Lesson 4 - it already ran every frame, today we just made one number depend on the wave.



Student's Choice

Pick ONE. Good-faith effort is a complete finish.

10 min

This Week's Twist

10 min

Level 3 works differently this week: pick ONE of five real capstone-level ideas.

- Destructible shields
- A power-up
- A bonus enemy
- Formation movement
- Title / game-over screen

These are open-ended on purpose. A good-faith attempt at the general idea - even if rough - is a complete Level 3 finish today. Pick whichever sounds the most fun to YOU.



Today's Exercise & Lab

Raising the Stakes - scale it, then choose your own.

5 min

Raising the Stakes

5 min

Scale enemy count and speed by wave, then pick one stretch feature - built on top of your existing project.

Level 1 - Minimal

Enemy count increases by one each wave (5, 6, 7).

MVP - a complete finish for today.

Level 2 - Complete

Speed ALSO scales; restart returns to wave 1's values.

Everything works end to end.

Level 3 - Refined

Choose ONE stretch feature; a good-faith attempt.

Ready to share.



Test a full 3-wave round when you're done, and confirm restarting brings everything back to wave 1's original feel.

Lab Time Notes

- Watch for the spawn-count formula updated in one place but not the enemies_remaining reset in the other - an easy one to half-do.
- 001 Game Creator students need extra time up front to correctly add and name enemy_6/enemy_7 before any scripting begins.
- If a cluster's wave 3 feels impossible rather than exciting, help them dial back their difficulty increments rather than pushing through.
- Level 3 is open-ended this week - help students scope their chosen feature down to something achievable in the remaining time.

The Stakes Are Raised!

More enemies, faster enemies, and a feature that's all your own.

Next week: a final polish pass, and the Semester 1 Capstone presentation.

Keep tuning your difficulty curve at home if you'd like - and keep building on your Level 3 feature if you're excited about it.